

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer from all the section as directed.

Section-A

Multiple choices questions

(Compulsory)

1. Choose the correct answer from the following
multiple choice questions : $2 \times 10 = 20$
 - (i) Who provides the interface to access the
services of the Operating system?

- (a) API
 - (b) System call
 - (c) Library
 - (d) Assembly instruction
- (ii) Where are placed the list of processes that prepared to be executed and waiting?
- (a) Job queue
 - (b) Ready queue
 - (c) Execution queue
 - (d) Process queue
- (iii) Which of the following Scheduling algorithms is preemptive scheduling?
- (a) FCFS Scheduling
 - (b) SJF Scheduling
 - (c) Network scheduling
 - (d) SRTF scheduling

(iv) What type of memory stores data in a swap file on a hard drive?

(a) Secondary memory

(b) Virtual memory

(c) Low memory

(d) RAM

(v) Which method is the best among file allocation methods?

(a) Linked

(b) Contiguous

(c) Indexed

(d) None of these

(vi) Which of the following scheduling reduces process flow time?

(a) FCFS

(b) SJF

- (c) LIFO
 - (d) All of these
- (vii) The address of the next instruction to be executed by the current process is provided by the :
- (a) CPU register
 - (b) Program counter
 - (c) Process stack
 - (d) Pipe
- (viii) What else is a command interpreter called :
- (a) Prompt
 - (b) Kernel
 - (c) Shell
 - (d) Command
- (ix) If the page size increases, the internal fragmentation is also _____ ?

- (a) Decreases
 - (b) Increases
 - (c) Remain constant
 - (d) None of these
- (x) Physical memory is broken into fixed-sized blocks called :
- (a) Frames
 - (b) Pages
 - (c) Backing Store
 - (d) None of these

Section-B

Short answer type questions

Answer any **four** questions of the following :

$$5 \times 4 = 20$$

- ✓ 2. Explain the process states with a diagram.
- ✓ 3. Differentiate between process and program.

4. Explain memory allocation strategies with example.
5. Explain the purposes of an operating system.
- ✓ 6. What is scheduler? Explain the types of scheduler.
7. Explain process control Block.
8. What is thrashing? Explain the causes of thrashing.
- ✓ 9. What do you mean by real time system?

Section-C

Long answer type questions

Answer any **two** questions of the following :

15×2=30

- ✓ 10. Define Operating system. Explain its functions.
11. What is file? Explain different file allocation methods.

✓ 12. What is paging ? Explain simple paging and demand paging.

✓ 13. For the processes listed below. Draw a chart and find average turnaround time and average waiting time for the following CPU scheduling algorithms.

(a) FCFS

(b) SJF

(c) SRTF

Process	Arrival time	Processing time
A	0	3
B	1	6
C	4	4
D	6	2

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